Which Keypoints Should ORB Keep?
Learning Classical Keypoint Selection from Image-Pair Labels

Anonymous submission

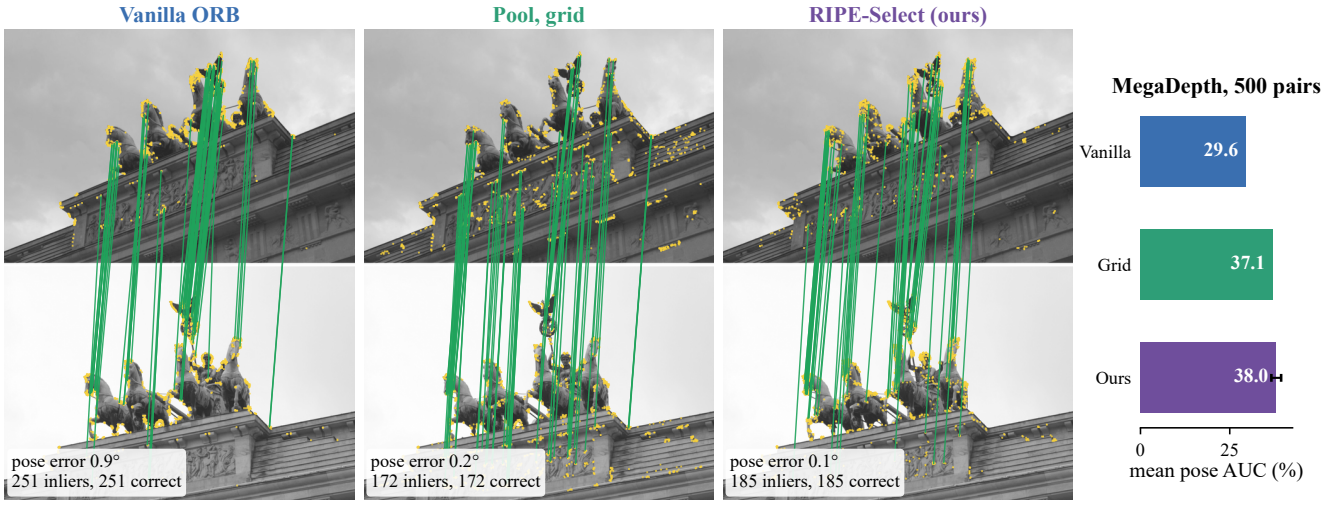


Figure 1. **Same detector, same descriptor, same budget—different keypoints.** ORB on a MegaDepth pair, keeping $K{=}2048$ keypoints (yellow) with OpenCV's response ranking, a spatial grid, or our learned policy. Lines are a random subset of 70 RANSAC inliers, green if geometrically correct and orange otherwise. The pair is not hand-picked: its gain is the closest to the mean gain over all 500 test pairs. Right: mean pose AUC over the benchmark (RIPE-SELECT: five seeds).

Abstract

Since their introduction, classical detectors such as ORB and BRISK have decided which keypoints to keep the same way: the K with the highest corner response. We show that this choice is worth learning. Response barely predicts which keypoints will match, and a small policy trained only with RIPE's image-pair reward—a kept keypoint scores when its match survives RANSAC on a same-scene pair and is penalised on a different-scene pair—picks much better ones. RIPE-SELECT *leaves the detector, the binary descriptor, the matcher and RANSAC untouched; it only re-ranks the detector's own candidates. Trained on two scenes on a laptop CPU, it raises MegaDepth-1500 pose AUC from 29.6 to* ***38.0*** *for ORB (+8.5, AUC@5°* ***+31 %*** *relative) and from 41.5 to* ***46.0*** *for BRISK. Every one of five training seeds improves significantly, and ORB with learned selection overtakes a learned binary feature reported in the literature. Analysing the policy, we find that it swaps two-thirds of the kept keypoints, ignores the detector response, and gains pose accuracy not by finding more correct matches but by spreading them over the image—a lesson it learns from a per-keypoint reward without ever seeing image coordinates.*

1. Introduction

Learned local features [**?** **?** **?** **?**] have advanced image matching rapidly, yet classical features remain what a large share of deployed systems run. ORB [**?**] is the front end of ORB-SLAM [**?**]; binary features are the default wherever compute, memory, power or licensing are tight, and even recent "fast" learned features [**?** **?**] are benchmarked against them. All of these pipelines share one step that has not changed since they were proposed: detect many candidates, then keep the K with the highest detector response.

Response is the wrong criterion. Response measures local contrast, not *matchability*. On MegaDepth, response separates candidates that end in a correct match from those that do not with a ROC-AUC of only 0.56–0.60, the correct-match rate barely falls from the strongest to the weakest response quartile, and a top-K cut by response throws away more than half of the correct matches present among the

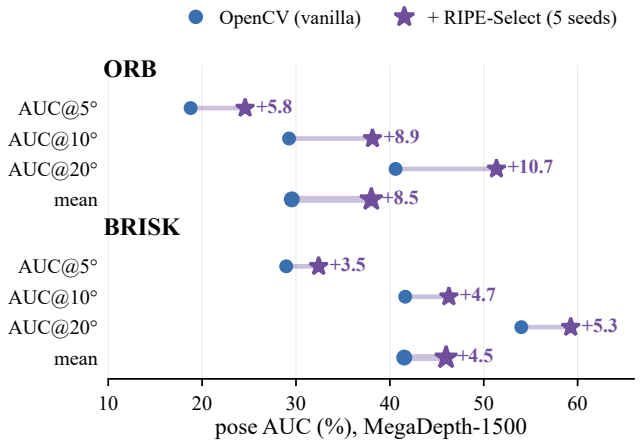


Figure 2. **Same detector, same descriptor, better keypoints.** Pose AUC on MegaDepth-1500 for OpenCV's response ranking and for RIPE-SELECT selection from the detector's own candidates (mean of five seeds); labels: gain in AUC points.

candidates (Fig. 9). Which K keypoints to keep is therefore an open, cheap and so far unexploited lever.

Learning the choice without labels. RIPE [**?**] trains a keypoint detector by reinforcement learning from nothing more than whether two images show the same scene. A classical detector has no heatmap to sample from, so we learn a *selection policy* over its candidates instead (Fig. 3). A small MLP scores every candidate from local cues and its descriptor; a budget-exact Bernoulli selection keeps exactly K in expectation; RANSAC on same-scene and different-scene pairs supplies the reward. Detector, descriptor, matcher and robust estimator are unchanged, so RIPE-SELECT is a drop-in module.

What we find. Under a controlled protocol in which every method shares images, candidates, budget, descriptor, matcher and RANSAC, learned selection improves ORB and BRISK at every pose threshold on MegaDepth-1500 (Fig. 2): +8.5 and +4.5 mean pose AUC, every seed significant. Opening the policy up, we find that it replaces about two-thirds of the response-ranked keypoints and that its gain comes from *where* the matches land more than from *how many* are correct: the policy discovers spatial spread on its own, the property that the grid bucketing of ORB-SLAM imposes by hand.

Contributions.

- RIPE-SELECT, a keypoint selection policy for classical detectors trained with RIPE's pair-label reward, with a budget-exact Bernoulli selection that removes RIPE's keypoint-count regulariser (Sec. 3).
- A controlled evaluation with five seeds, paired tests and published reference numbers, showing +8.5 (ORB) and +4.5 (BRISK) pose AUC over OpenCV on MegaDepth-1500 (Sec. 5).
- An analysis of what the policy learns—which cues it uses, which keypoints it swaps and why the swap helps—and how it compares with the spatial grid (Sec. 6).

2. Related work

Classical features. SIFT [**?**] detects difference-of-Gaussian extrema; ORB [**?**] combines oriented FAST [**?**] corners, ranked by a Harris score, with rotated BRIEF; BRISK [**?**] uses AGAST [**?**] corners and a 512-bit sampling-pattern descriptor. All three control the keypoint count with a response threshold and a top-K cut, which is the step we replace.

Selection and spatial distribution. Adaptive non-maximal suppression [**?** **?**] keeps strong keypoints that are far apart; ORB-SLAM [**?**] buckets keypoints into a grid. These rules decide *where* keypoints are kept from hand-set geometry. We use grid bucketing as our strongest baseline.

Predicting matchability. Hartmann *et al.* [**?**] classify which SIFT keypoints will match, supervised by structure-from-motion tracks. Our policy needs only image-pair labels and is trained on the outcome of matching and RANSAC itself; we also compare it with a supervised model trained on RANSAC labels.

Learned and efficient features. SuperPoint [**?**], Key.Net [**?**], DISK [**?**], ALIKED [**?**], SiLK [**?**] and DeDoDe [**?**] learn detectors from depth, pose or homography supervision; DISK and RIPE [**?**] use policy gradients, RIPE with only pair labels. ZippyPoint [**?**] and XFeat [**?**] target the speed regime of classical features with learned binary or compact descriptors. All of these replace the feature; we keep the classical one and learn only which keypoints to keep.

3. Method

Candidate pool. We run the unchanged OpenCV detector with lowered thresholds—FAST threshold 7 with up to 8K points for ORB, AGAST threshold 10 instead of 30 for BRISK—and keep the $4K$ strongest candidates by response. The unchanged OpenCV descriptor is computed for every candidate.

Cues. Each candidate i is described by $\mathbf{f}_i$: 20 scalar cues followed by its descriptor bits (276 inputs for ORB, 532 for BRISK). The cues cover response and its rank, scale and pyramid level, near-duplicate detections, local density and rank in the neighbourhood, intra-image descriptor distinctiveness, and patch statistics (full list in Sec. C). Image coordinates are deliberately excluded, so the policy cannot memorise scene-specific locations—and cannot be told directly to spread its keypoints.

Budget-exact selection. An MLP ($\to 128 \to 64 \to 1$, ReLU) maps $\mathbf{f}_i$ to a score s_i. During training candidate i is

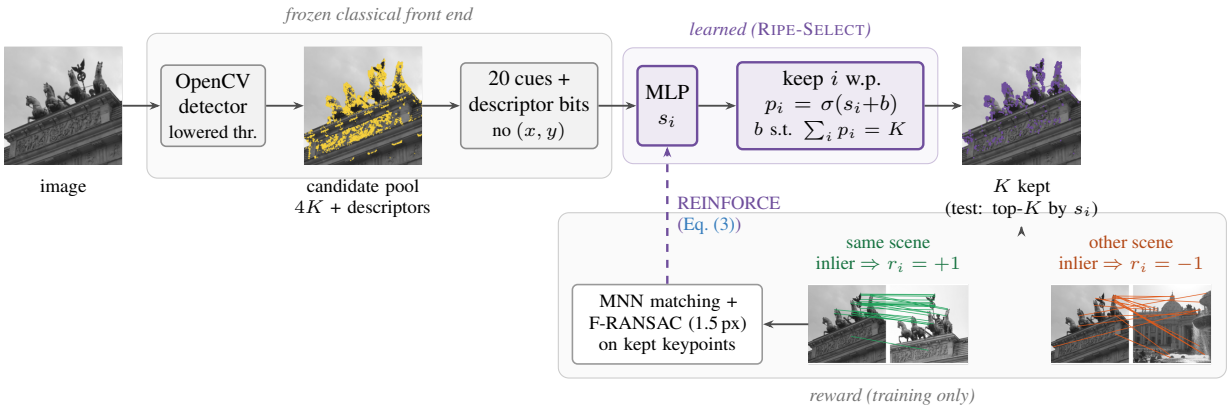


Figure 3. **RIPE-SELECT.** A classical detector run with lowered thresholds proposes a candidate pool; an MLP scores every candidate from 20 local cues and its descriptor, without image coordinates. During training each candidate is kept by a Bernoulli draw whose per-image bias b fixes the expected count to the budget K; kept keypoints are matched and earn $+1$ when they survive RANSAC on a same-scene pair and -1 on a different-scene pair. At test time the top-K scores are kept and passed to the unchanged pipeline.

kept, $a_i{=}1$, with probability

$$p_i = \sigma(s_i + b), \qquad b \text{ such that } \textstyle\sum_i p_i = K. \tag{1}$$

Since $\sum_i p_i$ is monotone in b, b is found per image by bisection. The expected number of kept keypoints is thus fixed to the budget: the policy can change *which* keypoints are kept but not *how many*, which removes RIPE's keypoint-count regulariser and the quantity–quality trade-off it controls. At test time we keep the top-K scores.

Reward. For a training pair with label $y \in \{\text{same}, \text{diff}\}$, kept keypoints are matched by mutual nearest neighbour (MNN) and a fundamental matrix is fitted with RANSAC [**?**] at 1.5 px on 1200 px images. Following RIPE, a kept keypoint receives

$$r_i = \begin{cases} +1 & \text{inlier},\ y = \text{same} \\ -1 & \text{inlier},\ y = \text{diff} \\ 0 & \text{otherwise}, \end{cases} \tag{2}$$

so negative pairs penalise keypoints that produce confident but meaningless matches.

Training. We maximise the expected reward with REINFORCE [**?**],

$$\nabla_\theta J = \mathbb{E}\Big[\textstyle\sum_i (r_i - \bar{r})\, \nabla_\theta \log \pi_\theta(a_i \mid \mathbf{f}_i)\Big], \tag{3}$$

with the per-image mean reward $\bar{r}$ as baseline. One step is summarised in Algorithm 1.

Algorithm 1: one training step of RIPE-SELECT

Input: 12 image pairs (50 % different-scene), policy π_θ, budget K

1: **for** each image I in the batch **do**
2: pool ← top-$4K$ detections; $s \leftarrow \pi_\theta(\mathbf{f})$
3: solve b with $\sum_i \sigma(s_i{+}b) = K$ — Eq. (1)
4: sample $a_i \sim \text{Bernoulli}(\sigma(s_i{+}b))$
5: **for** each pair **do** MNN + F-RANSAC on kept keypoints
6: assign r_i to kept keypoints by pair label — Eq. (2)
7: $\bar{r} \leftarrow$ per-image mean of r; update θ with Adam — Eq. (3)

4. Experimental setup

Training. 350 images each from the IMC-2020 [**?**] scenes *Palace of Westminster* and *Grand Place Brussels*; positive pairs are covisible pairs from the COLMAP models, negative pairs are drawn across scenes. 1200 steps of 12 pairs, Adam (10^{-3}), 40–90 minutes per run on a laptop CPU. Checkpoints are selected on the held-out scene *Reichstag*; no test image is used for training or selection (hyper-parameters in Sec. D).

Benchmarks. **MegaDepth-1500** [**?** **?**]: relative pose on scenes 0015 and 0022, AUC of the pose error at 5/10/20°, $K{=}2048$, 500 evenly spaced pairs covering both scenes. **HPatches** [**?**]: homography on 540 pairs, AUC of the corner error at 1/3/5 px, $K{=}1024$. Both follow RIPE's protocol [**?**] (1200 px for MegaDepth, 480 px for HPatches, MNN, PoseLib), so its published numbers serve as a reference.

Controlled protocol. Within a table every row shares the image, candidate pool, budget, OpenCV descriptor, MNN matcher and PoseLib RANSAC [**?**] (100k iterations, fixed seed). The RANSAC threshold is tuned for vanilla OpenCV and reused for all rows, which favours the baseline.

Baselines. *Vanilla*: OpenCV defaults. *Pool, top-K response*: our candidate pool ranked by response. *Pool, grid*: the same pool in a $16{\times}16$ grid, taken round-robin by response as in ORB-SLAM. *Supervised*: our network trained with binary cross-entropy on the RANSAC inlier labels of

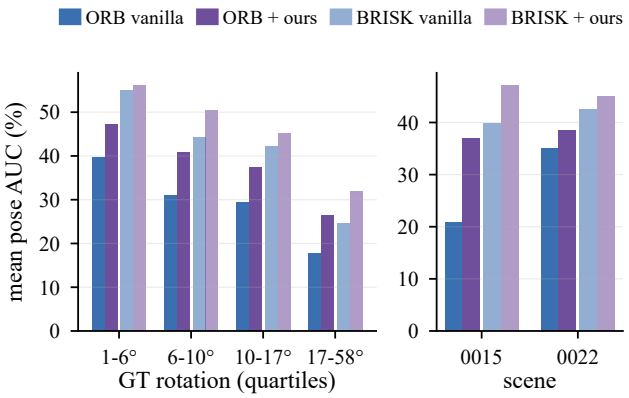


Figure 4. **Gains across difficulty.** Mean pose AUC on MegaDepth by quartile of the ground-truth rotation between the two views (left) and by scene (right); RIPE-SELECT: mean of five seeds.

the full pool, which isolates the effect of RL.

Statistics. RIPE-SELECT rows are means over five seeds ($\pm$ sample std). Per seed, a paired t-test over the 500 test pairs compares RIPE-SELECT with vanilla; across seeds, a one-sample t-test on the five gains. Ablations use a paired bootstrap over test pairs (2000 resamples, 95 % CI) with the seed-0 policy.

5. Results

MegaDepth. Learned selection lifts both detectors by a wide margin (Tab. 1, top): ORB from 29.6 to 38.0 mean AUC and BRISK from 41.5 to 46.0, with the largest relative gain at the strictest threshold (ORB AUC@5° 18.8 $\rightarrow$ 24.6, +31 %), all while keeping the binary descriptors. The improvement holds at every threshold from 1° to 30° (Fig. 7).

Reliability. The gain is not a lucky seed (Fig. 6a). All five seeds improve on vanilla—ORB by +6.2 to +9.8, BRISK by +3.4 to +5.8—each significantly over the 500 pairs (paired t-test, $p \leq 0.016$), and the gain is significant across seeds (one-sample t-test, n=5: ORB t=13.9, BRISK t=10.9, both $p < 0.001$; 95 % CI ORB 8.5 ± 1.7, BRISK 4.5 ± 1.1). Per pair, RIPE-SELECT improves 54 % of the ORB pairs by more than half a point and worsens 28 % (BRISK: 48 % and 38 %; Fig. 6b): the gain is broad, not carried by a few pairs.

Where does it help? Figure 4 splits the 500 test pairs by the ground-truth rotation between the two views. ORB gains in every quartile, from +7.5 on near-identical views (1–6°) to +8.7 on the widest baselines (17–58°). For BRISK the gain grows with the viewpoint change, from +1.2 to +7.3: the harder the pair, the more it matters which keypoints are kept. Both scenes improve, most strongly the cathedral of scene 0015, where ORB rises from 20.9 to 37.0.

Qualitative results. Figure 5 shows ORB pairs at the 95th, 80th and 60th percentile of the per-pair gain, selected by rank rather than by eye. In the first, OpenCV's keypoints produce 17 RANSAC inliers, none of them correct, and a pose that is off by 149°; with RIPE-SELECT, 47 of 49 inliers are correct and the error drops to 2.4°. In the third, the response-ranked keypoints cluster on the quadriga and RANSAC locks onto a wrong solution (141°), while the learned selection adds keypoints on the gate and the building behind it and recovers the pose to 16.5°.

Against published binary features. Table 2 sets these numbers next to published results for binary features. Our vanilla ORB reproduces the ORB row reported by XFeat [**?**] to within 1.7 AUC at every threshold, so the protocols agree closely. With learned selection, ORB—a 2011 detector with a 256-bit descriptor—surpasses ZippyPoint [**?**], a learned binary feature, at every threshold, and BRISK with RIPE-SELECT is ahead of both by a wide margin.

HPatches. On planar HPatches (Tab. 1, bottom) RIPE-SELECT is on par with vanilla ($\pm$0.0 mean AUC for ORB, +0.3 for BRISK) and below the grid. These scenes are planar and well textured, so almost every textured keypoint is valid and the choice among them matters little; spreading them, which the grid does by construction, is what helps there. The policy was also trained at 1200 px on outdoor photographs, while HPatches images are about 480 px.

6. What does the policy learn?

Is there anything to learn? Yes. Figure 9 bins pool candidates by response quartile: on MegaDepth the share that becomes a correct match only falls from 8 % to 4–6 % between the strongest and the weakest quartile, and top-K by response keeps only 40–44 % of the pool's correct matches.

The policy rewrites the selection. Compared with top-K response on the same pool, RIPE-SELECT replaces 68 % of the kept keypoints for ORB and 64 % for BRISK (40 test images, Fig. 8). Its score is almost uncorrelated with response (ρ=0.09). It rises with patch brightness and contrast (ORB ρ=0.42, BRISK ρ=0.35) and, perhaps surprisingly, *falls* with intra-image descriptor distinctiveness (ρ between -0.36 and -0.60 for the distance to the nearest and second-nearest descriptor). A binary descriptor far from every other one in its image may be dominated by noise, which does not repeat in the second view; a plausible reading is that the policy prefers descriptors of structured, recurring texture. It also slightly avoids dense clusters ($\rho = -0.06$ and -0.11 with local density). Figure 10 shows the effect on one image: response peaks on the sculpture, while the learned score is spread over the cornice and facade.

How fast does it learn? Figure 11 tracks the validation score on the held-out *Reichstag* scene. For BRISK, all five seeds pass the response ranking (46.5) within the first 100 steps and settle at 58–60. For ORB, even the untrained policy (26–32), which keeps a nearly arbitrary subset of the larger pool, is already well above response ranking (17.2), another sign of how much response clusters ORB's keypoints; train-

	ORB				BRISK			
MegaDepth-1500, relative pose (K=2048)	@5°	@10°	@20°	Mean	@5°	@10°	@20°	Mean
Vanilla OpenCV	18.8	29.3	40.6	29.6	29.0	41.6	54.0	41.5
Pool, top-K response	15.4	25.2	36.5	25.7	30.8	43.9	56.3	43.7
Pool, grid	**25.3**	37.0	49.1	37.1	**33.8**	**48.7**	**62.1**	**48.2**
Supervised	20.1	32.3	45.4	32.6	33.4	48.3	61.7	47.8
RIPE-SELECT (5 seeds)	24.6	**38.1**	**51.4**	**38.0 ± 1.4**	32.4	46.3	59.3	46.0 ± 0.9
gain over vanilla	+5.8	+8.9	+10.7	+8.5	+3.5	+4.7	+5.3	+4.5
HPatches, homography (K=1024)	@1 px	@3 px	@5 px	Mean	@1 px	@3 px	@5 px	Mean
Vanilla OpenCV	21.0	45.0	56.9	41.0	27.5	58.0	69.4	51.6
Pool, top-K response	18.4	42.9	55.1	38.8	28.2	58.6	70.1	52.3
Pool, grid	**24.5**	**51.0**	**62.3**	**45.9**	**31.9**	**61.9**	**72.6**	**55.5**
Supervised	18.1	42.5	54.4	38.3	27.1	57.2	69.0	51.1
RIPE-SELECT (5 seeds)	20.4	45.3	57.1	40.9 ± 0.3	28.1	58.0	69.7	51.9 ± 0.1
gain over vanilla	-0.5	+0.3	+0.2	+0.0	+0.6	+0.0	+0.3	+0.3

Table 1. **Main results.** AUC (%) of the pose error on MegaDepth-1500 (500 pairs) and of the homography corner error on HPatches. Every row uses the same images, budget, descriptor, matcher and RANSAC; the pool rows, supervised and RIPE-SELECT select from the same candidate pool. RIPE-SELECT: mean over five seeds, ± std of the mean AUC. **Bold**: best, underlined: second best.

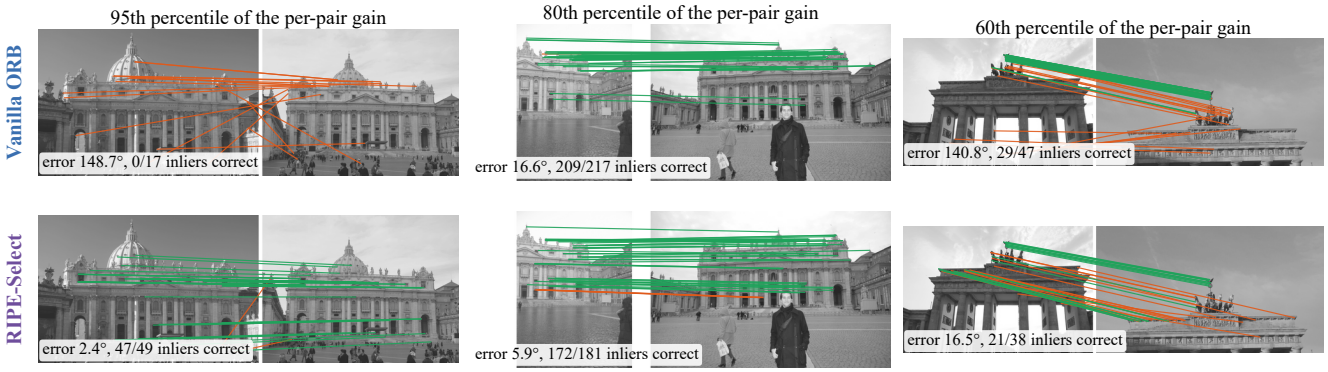


Figure 5. **Qualitative results** for ORB on MegaDepth, K=2048. Columns: pairs at the 95th, 80th and 60th percentile of the per-pair gain of RIPE-SELECT (seed 0), chosen by rank. Top: OpenCV's response ranking; bottom: RIPE-SELECT. Lines: a random subset of 50 RANSAC inliers, green if correct (symmetric epipolar error below 2 px), orange otherwise. Labels: pose error and correct/total inliers.

MegaDepth-1500 (binary features)	@5°	@10°	@20°	Mean
*Reported by XFeat [**?**]*				
ORB [**?**]	17.9	27.6	39.0	28.2
ZippyPoint [**?**]	23.6	34.9	46.3	34.9
Ours (top-2048, MNN, PoseLib)				
ORB (OpenCV)	18.8	29.3	40.6	29.6
+ RIPE-SELECT	24.6	38.1	51.4	38.0
BRISK (OpenCV)	29.0	41.6	54.0	41.5
+ RIPE-SELECT	**32.4**	**46.3**	**59.3**	**46.0**

Table 2. **Binary features on MegaDepth-1500,** pose AUC (%). XFeat's rows use all 1500 pairs, 4096 keypoints and LO-RANSAC on the essential matrix; ours use 500 evenly spaced pairs. **Bold**: best, underlined: second best.

ing then lifts the best checkpoints to 34–36. Runs take 40–90 minutes on a laptop CPU.

More correct matches, or better-placed ones? Table 3 separates the two. For BRISK, RIPE-SELECT raises both: match precision (MMA@3 px 24.7 % → 26.9 %), inlier ratio (22.2 % → 24.5 %) and the number of positive-pair inliers (100 → 125). For ORB, precision and inlier ratio are unchanged and the policy keeps *fewer* RANSAC inliers than vanilla (137 *vs.* 156); the keypoints it adds are, one by one, no more likely to match than those it drops (6.7 % *vs.* 8.1 % end in a correct match). Yet ORB gains the most pose accuracy. The difference is placement: the inliers of RIPE-SELECT cover 9.7 % of the 16×16 image cells, against 7.6 % for vanilla and 7.0 % for top-K response of the same pool

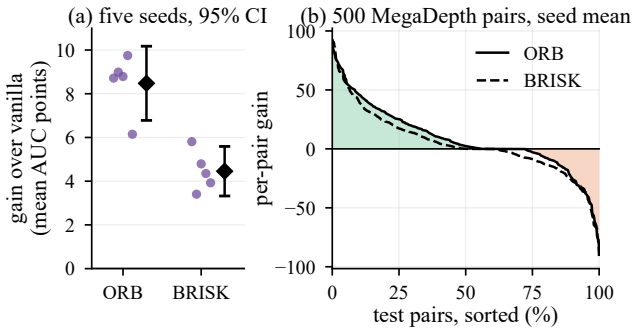


Figure 6. **The gain is consistent.** (a) Gain in mean pose AUC over vanilla for each of five training seeds; black: mean and 95 % CI. (b) Per-pair gain (contribution to mean AUC, averaged over seeds), sorted; green/orange: ORB pairs that improve / worsen.

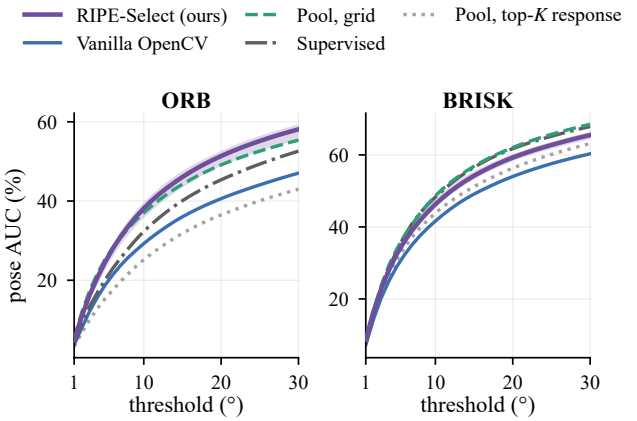


Figure 7. **Pose AUC vs. error threshold** on MegaDepth. Band: range over five RIPE-SELECT seeds.

(Fig. 12). Spread-out correspondences constrain the relative pose far better than the same number clustered on one textured object.

Finding. Trained with a per-keypoint reward and without image coordinates, the policy learns to spread its matches over the image. For ORB this, not higher per-keypoint precision, is where the pose gain comes from.

The grid takes the same route further. Bucketing the pool into a grid spreads inliers even more (12.3 % of cells for ORB, 17.0 % for BRISK) at the price of precision (ORB inlier ratio 24.0 % → 17.6 %), and it reaches a similar pose accuracy: RIPE-SELECT is ahead by 0.9 for ORB and behind by 2.2 for BRISK, neither difference significant (Fig. 13). Forcing the learned scores into the same grid does not combine the two: it lowers ORB to 34.0, because the grid then overrides the placement the policy chose. Learning precision and coverage *jointly*—a coverage-aware reward, or a selection rule that trades the two—is the clear next step.

Is it just the larger pool? No. Ranking the same pool by response *lowers* ORB accuracy by 3.9 AUC: the extra low-

		MMA (%)			Inl.	Cov.	Time
		@1 px	@3 px	@5 px	(%)	(%)	(ms)
ORB	Vanilla	16.9	27.4	30.0	**24.0**	7.6	**20**
	Pool, top-K	16.5	27.1	29.9	23.9	7.0	70
	Pool, grid	12.2	20.1	22.2	17.6	**12.3**	70
	RIPE-SELECT	**17.0**	**27.5**	**30.1**	**24.0**	9.7	838
BRISK	Vanilla	16.7	24.7	27.1	22.2	10.6	**81**
	Pool, top-K	16.1	24.1	26.3	21.5	12.9	295
	Pool, grid	15.0	22.3	24.3	20.2	**17.0**	295
	RIPE-SELECT	**18.1**	**26.9**	**29.2**	**24.5**	14.6	1077

Table 3. **Quality, coverage and cost** on MegaDepth (seed 0). MMA: share of MNN matches with epipolar error below the threshold. Inl.: RANSAC inlier ratio. Cov.: share of 16×16 cells containing a RANSAC inlier (50 pairs). Time: median single-thread detection and selection per image, measured in isolation.

threshold candidates pile up on the most textured regions. RIPE-SELECT (seed 0) beats this pool top-K by +12.6 for ORB and +3.7 for BRISK, both significant (Fig. 13).

Does reinforcement learning matter? For ORB, RIPE-SELECT beats the supervised model trained on the same RANSAC pseudo-labels by +5.7 [+3.1, +8.2]; for BRISK the two are equal (−0.5 [−2.9, +1.9]). The negative-pair penalty is not the explanation: on 50 MegaDepth pairs each also matched against an image of the other scene, RIPE-SELECT lowers the mean number of ORB inliers on these negative pairs only from 21.8 to 19.5, and the ratio of positive to negative inliers is unchanged (7.0 *vs.* 7.1). A more likely cause is the training signal itself: RL is rewarded for matches among *sampled* subsets of the pool, exactly as at test time, so it can learn which keypoints work *together*—including where they lie—whereas the supervised labels judge each candidate in isolation on the full pool.

What does it cost? The policy needs the larger pool and the cues. In an isolated single-thread benchmark (median of 20 MegaDepth images) the pipeline takes 838 ms per image for ORB and 1077 ms for BRISK, against 20 and 81 ms for OpenCV (Fig. 14). About 740 ms of this is cue computation in unoptimised Python; the MLP takes 27–41 ms and the final selection about 1 ms. Most cues are box filters and nearest-neighbour counts that map directly to C++ or a GPU.

7. Limitations

Training uses two scenes (700 images), no hyper-parameter search and one validation scene; RIPE uses the DISK MegaDepth subset and about three GPU-days, so more training data is a natural extension. MegaDepth results use 500 of the 1500 pairs, and the ablations and bootstrap tests use seed 0. Cue computation in Python makes selection slow. We focus on ORB and BRISK, the binary detectors behind most lightweight pipelines; applying RIPE-SELECT to float-descriptor detectors such as SIFT and SURF is ongoing

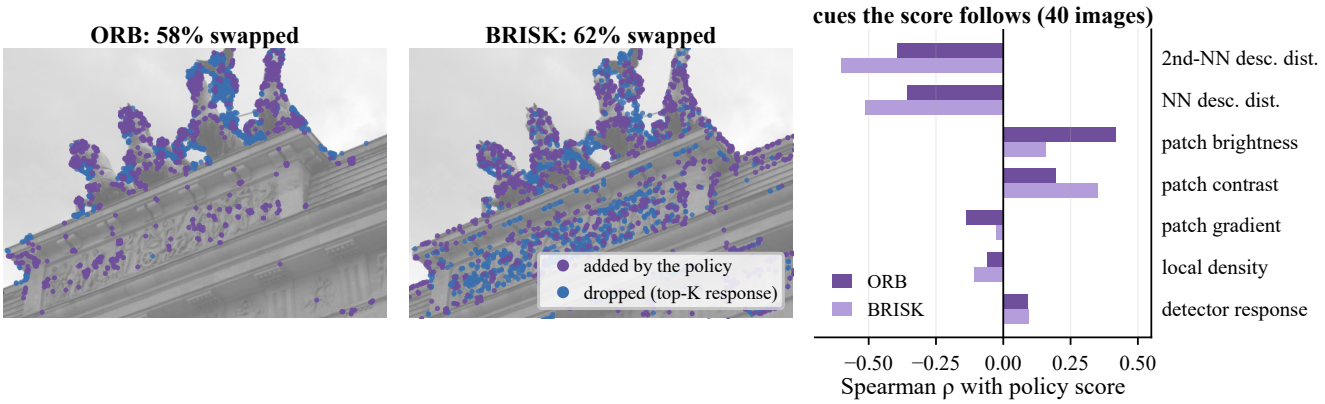


Figure 8. **The policy rewrites the selection.** Left: keypoints that RIPE-SELECT keeps but top-K response of the same pool does not (purple), and the ones it drops (blue), on the first image of Fig. 1 (zoomed). Response crowds the sculpture; the policy moves keypoints down onto the cornice and the facade. Right: Spearman correlation between the policy score and individual cues within an image, averaged over 40 MegaDepth test images. The score follows patch appearance and descriptor similarity; it is almost independent of the detector response that OpenCV ranks by.

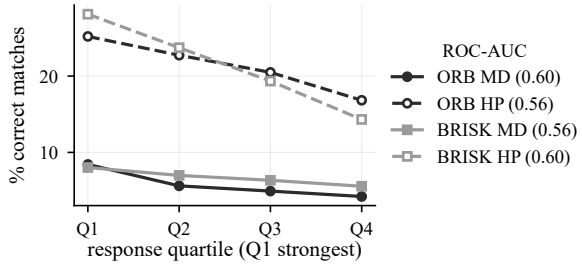


Figure 9. **Response is a weak cue.** Share of candidates that become correct matches, by response quartile (MD: MegaDepth, HP: HPatches); legend: ROC-AUC of response as a predictor.

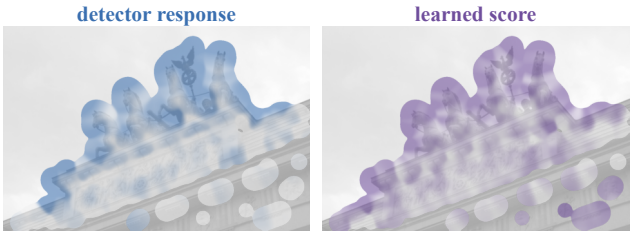


Figure 10. **Response map vs. learned score map** over the ORB candidate pool of the first image of Fig. 1: within-image rank of each candidate, smoothed (darker = ranked higher).

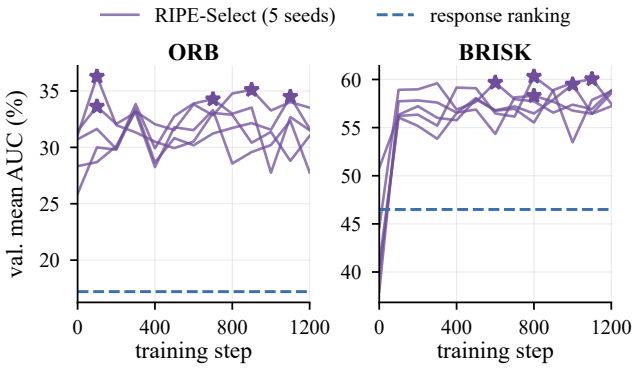


Figure 11. **Training curves.** Validation mean AUC on the held-out *Reichstag* scene every 100 steps, five seeds; stars: selected checkpoints. Dashed: top-K response ranking on the same scene.

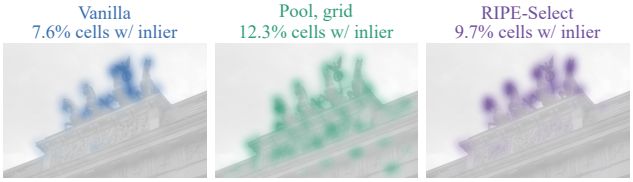


Figure 12. **Where the kept keypoints are.** Density of the 2048 kept ORB keypoints on the first image of Fig. 1. Titles: share of 16×16 cells containing a RANSAC inlier, averaged over 50 MegaDepth pairs.

work.

8. Conclusion

Which keypoints a classical detector keeps is worth learning, and it can be learned from image-pair labels alone. Without touching detector, descriptor or matcher, RIPE-SELECT lifts ORB by 8.5 and BRISK by 4.5 pose AUC on MegaDepth-1500, consistently across seeds, and moves ORB past a learned binary feature. The policy discards the detector's response ranking, rewrites two-thirds of the selection and, without seeing coordinates, learns to spread its matches—the same property a hand-made grid enforces. Teaching the policy to trade precision against coverage explicitly, and making the cues fast, are the steps that would turn this into a free

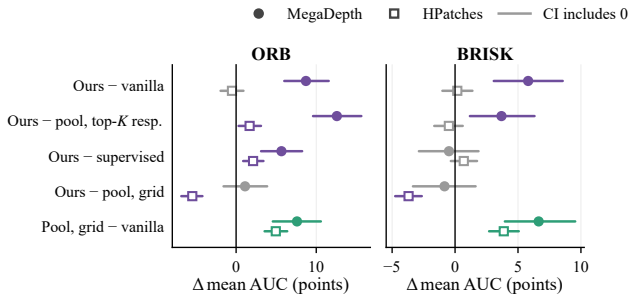


Figure 13. **Paired bootstrap** differences in mean AUC with 95 % CIs (seed-0 policy). Grey: the interval includes zero.

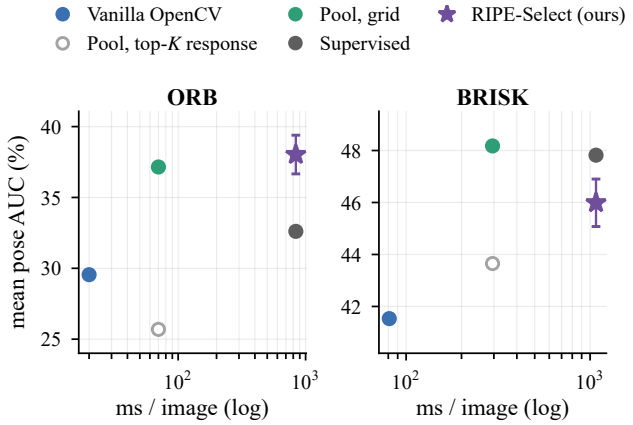


Figure 14. **Accuracy vs. cost** on MegaDepth. RIPE-SELECT: five-seed mean ± std; supervised shares its runtime.

upgrade for every ORB-based system.

Supplementary Material

Which Keypoints Should ORB Keep?

A. Per-seed results

		MegaDepth AUC			HPatches AUC			Val.
		@5°	@10°	@20°	@1 px	@3 px	@5 px	AUC
ORB	seed 0	25.3	38.2	51.3	20.3	44.8	56.3	35.1
	seed 1	25.0	38.6	52.0	20.3	45.2	57.3	34.3
	seed 2	24.5	38.5	52.0	20.7	45.6	57.3	34.5
	seed 3	25.6	39.7	52.6	20.4	45.3	56.9	36.3
	seed 4	22.6	35.7	48.8	20.5	45.8	57.7	33.6
BRISK	seed 0	33.9	47.9	60.2	28.3	57.9	69.2	59.5
	seed 1	31.3	45.3	58.2	27.8	58.2	70.2	60.0
	seed 2	32.6	46.7	59.7	28.3	57.9	69.5	58.3
	seed 3	32.6	46.2	58.8	27.8	58.0	69.9	60.3
	seed 4	31.7	45.4	59.3	28.1	58.1	69.6	59.7

Table 4. RIPE-SELECT per training seed. Val.: best mean AUC on the held-out *Reichstag* scene, used for checkpoint selection (plain response: ORB 17.2, BRISK 46.5).

B. Additional qualitative results

Figure 15 shows the BRISK pair whose per-pair gain is closest to the BRISK mean gain; Fig. 16 shows an ORB pair on which RIPE-SELECT is worse than vanilla, chosen as the pair whose loss is closest to the mean ORB gain with opposite sign. In the failure case all three selections recover the pose roughly (8.5–10.6°), and RIPE-SELECT has more correct inliers but concentrated in a smaller region.

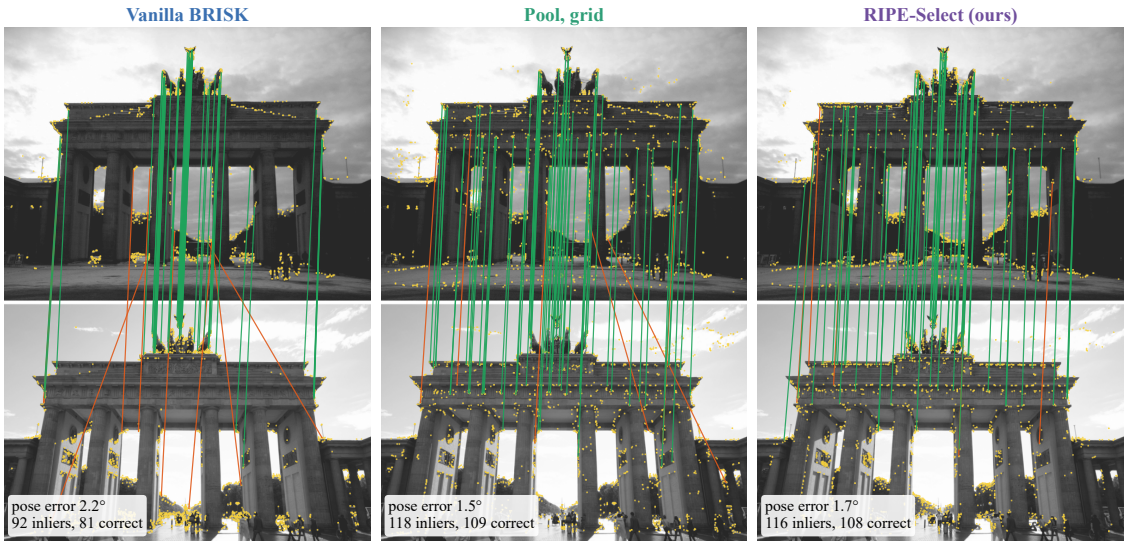


Figure 15. BRISK, representative pair (per-pair gain 5.8 *vs*. mean 5.8 AUC points).

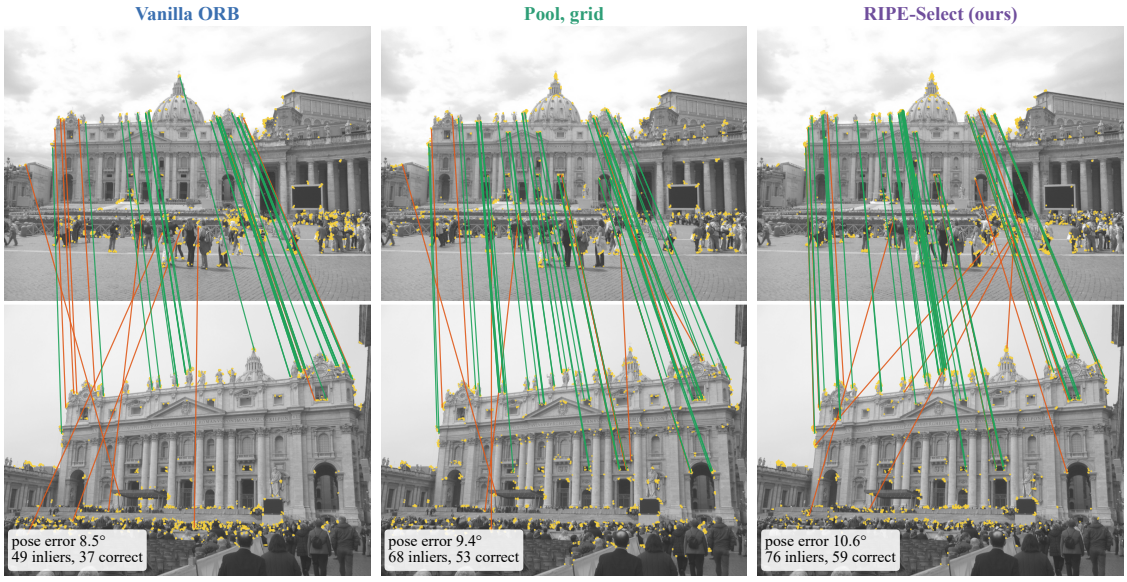


Figure 16. ORB, failure case (per-pair change -8.6 AUC points).

C. Per-candidate cues

The 20 scalar cues computed for every pool candidate: (1) log response; (2) log response standardised over the image; (3) response rank in the pool; (4) log size; (5) pyramid octave; (6) layer within the octave (0 for ORB and BRISK); (7) number of other detections within 0.5 px; (8) number of candidates within 16 px; (9) response rank among those neighbours; (10, 11) distance to the nearest and second-nearest other descriptor in the same image, normalised by the image median; (12) their ratio; (13–15) mean intensity, intensity standard deviation and mean gradient magnitude in a box of radius $0.75\times$ size; (16, 17) mean and standard deviation relative to the whole image; (18) log pool size; (19, 20) image mean intensity and standard deviation. They are followed by the descriptor as ± 1 bits (256 for ORB, 512 for BRISK).

D. Hyper-parameters and protocol

Pool	$4K$ candidates (8192 at $K{=}2048$); ORB: FAST thr. 7, up to 8K; BRISK: AGAST thr. 10 (default 30)
Policy	MLP, inputs $\to$ 128 $\to$ 64 $\to$ 1, ReLU
Training	1200 steps $\times$ 12 pairs, 50 % negative, $K{=}2048$, Adam with learning rate 10^{-3}
Reward RANSAC	fundamental matrix, 1.5 px at 1200 px, 2000 iterations
REINFORCE baseline	per-image mean reward
Validation	every 100 steps, 96 pairs of *Reichstag*
Test RANSAC	PoseLib, 100k iterations, seed 0
RANSAC threshold	tuned on vanilla: MegaDepth 1.0 px (from 1.0, 1.5); HPatches ORB 2 px, BRISK 4 px (from 1, 2, 3, 4, 6)
Runtime	single thread, warm-up pass, median over 20 images (`timing_benchmark.py`)
Policy analysis	seed-0 policies; 40 MegaDepth test images for cue correlations, 50 pairs for match rates and coverage

Table 5. Training and evaluation settings.